

CS & IT ENGINEERING



Operating System

Memory Management

Lecture -4

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Recap of Previous Lecture



Topic

TLB Mapping

Topic

Segmentation

Topics to be Covered



Topic

Virtual Memory

Topic

Effective Memory Access Time





Topic : Virtual Memory

- Feature of OS
- Enables to run larger process with smaller available memory





Topic : Virtual Memory

Process

V/I bit

000	000		0
001	001	2	0
010	010	3	1
011	011		0
100	100	0	1
101	101	2	1
110	110		0
111	111	1	1

8 pages

mem

0 00	Page 4
1 01	Page 7
2 10	page $\frac{1}{5}$
3 11	Page 2

4 frames

secondary mem.





Topic : Demand Paging

- **Demand Paging:**

Bring pages in memory only when CPU demands

- **Pure demand paging:-**
when process starts then it's no^{any} page in mm

- **Page Fault:**

When the demanded page is not available in physical memory

- OS provides page fault service $\Rightarrow$ OS brings faulted page from disk to mm and updates page table accordingly.
- $\rightarrow$ After p.f. service, the instruction which caused page fault will restart.



Topic : How to Check Page Hit/Fault

↓
by checking valid / Invalid bit





Topic : Saving Page Swap Time

	f	$\frac{v}{i}$	Dirty/modified bit
0			
1			
2			

Dirty/modified bit { $0 \Rightarrow$ Page had only read
 $1 \Rightarrow$ Page had write also

only dirty page are written
back to disk while replacement



Topic : Effective Memory Access Time

assume page fault rate = p
page hit rate = $1-p$

$$E.M.A.T. = (1-p) * \text{mem access time w/o page fault} + p * \text{mem access time for page fault}$$



Topic : Question

GATE-2018



Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory, and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process.

(A) $(D - M) / (X - M)$

✓ (B) $(X - M) / (D - M)$

(C) $(D - X) / (D - M)$

(D) $(X - M) / (D - X)$

$$X = (1-p)M + p * D$$

$$X = M - pM + pD$$

$$X - M = (D - M)p$$

$$\left(\frac{X - M}{D - M} \right) = p$$

CPU generates L.A.

↓
search P.T. in mm

hit



P.A



access mm
to get content

fault



p.f. service Time

$$E_{MAT} = t_{man} + (1-p) * t_{mm} + p * p.f. \text{ service time}$$



Topic : Question



mm access time = 200 n sec

Page fault rate = 1 %

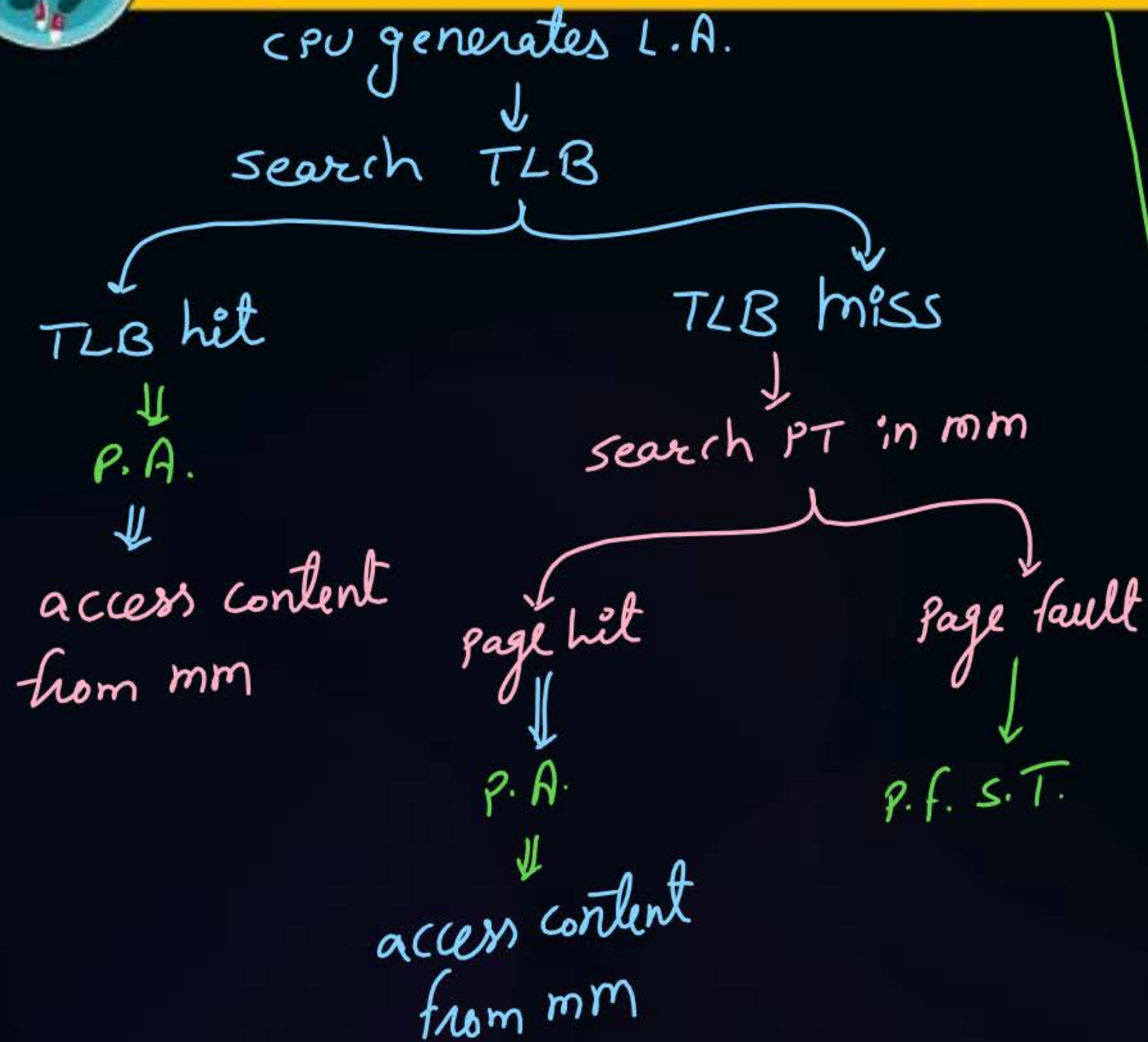
p.f. service time = 100 micro sec

E. M. A. T. = 1398 nanoseconds?

$$\begin{aligned} &= 200 + (0.99 * 200) + 0.01 * 100 * 1000 \text{ ns} \\ &= 1398 \text{ ns} \end{aligned}$$



Topic : Effective Memory Access Time with TLB



$$EMAT = t_{TLB} + H * t_{mm}$$

$$+ (1-H) \left[t_{mm} + (1-p) t_{mm} + p * pfst \right]$$



Topic : Question

TLB hit = 90%

TLB access time = 10n sec

mm access time = 200n sec

Page fault rate = 1 %

p.f. service time = 100 micro sec

E. M. A. T. = 329.8 ns?

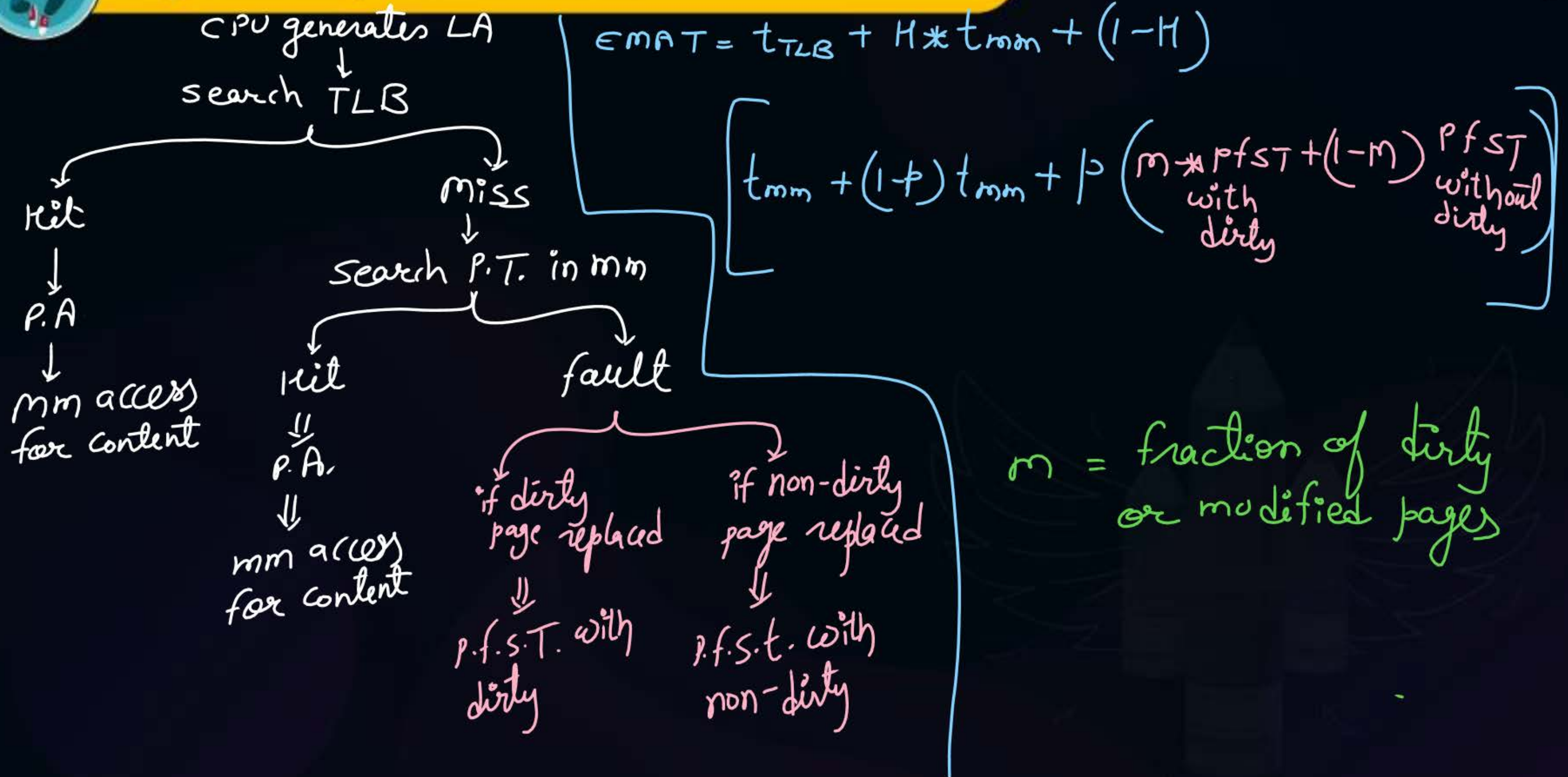
$$EMAT = 10 + 0.9 * 200$$

$$+ 0.1 \left[200 + 0.99 * 200 + 0.01 * 100000 \right]$$

$$= 329.8 \text{ ns}$$



Topic : Dirty Bit with TLB





Topic : Question

$$H = 85\%$$

$$t_{TLB} = 20 \text{ n sec}$$

$$t_{mm} = 200 \text{ n sec}$$

$$P = 1\%$$

$$M = 2\%$$

$$\text{P.F.S.T. with dirty page} = 5000 \mu \text{ sec}$$

$$\text{P.F.S.T. without dirty page} = 2000 \mu \text{ sec}$$

$$\text{E. M. A. T.} = \underline{3339.7 \text{ ns}}?$$

$$= 20 + 0.85 * 200 + 0.15 \left[200 + 0.99 * 200 + 0.01 * \left(0.02 * 5000000 + 0.98 * 2000000 \right) \right]$$
$$= 3339.7 \text{ ns}$$

P.F.S.T. without dirty page replacement

$$= \text{page transfer time from disk to mm} + \text{P.T. update time in mm}$$

P.F.S.T. for dirty page

$$\Rightarrow \text{page transfer time from mm to disk for dirty page} + \text{page transfer time from disk to mm for faulted page} + \text{P.T. update time in mm}$$



Topic : Question

TLB Hit = 90%

$t_{TLB} = 15 \text{ n sec}$

$T_{mm} = 120 \text{ n sec}$

$P = 2\%$

$M = 10\%$

Time taken to transfer a page b/w mm & sm = 10000 n sec

E.M.A.T. = 169 ?

$$p.f.s.T \text{ with dirty} = (2 * 10000) + 120 = 20120 \text{ ns}$$

$$p.f.s.T \text{ w/o dirty} = 10000 + 120 = 10120 \text{ ns}$$

$$\begin{aligned} &= 15 + 0.9 * 120 + 0.1 \left[120 + 0.98 * 120 + 0.02 \right. \\ &\quad \left. (0.9 * 10120 + 0.1 * 20120) \right] \\ &= 169 \text{ ns} \end{aligned}$$



Topic : Question

H.W.

[GATE-2020]

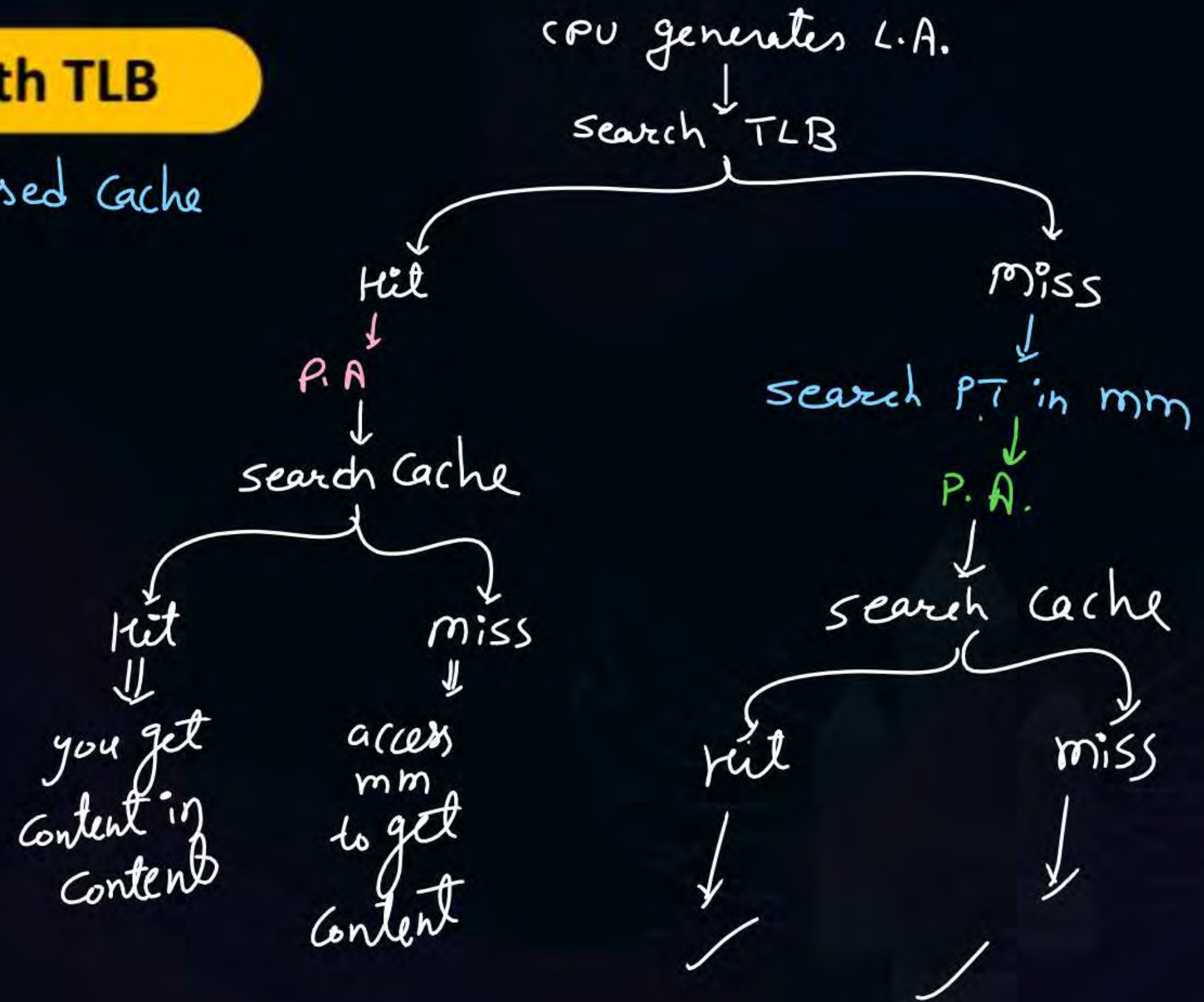


- #Q. Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is_____.



Topic : Cache with TLB

physically addressed Cache



Content access time using P.A. $(t_{CA}) = t_{cm} + (1-H) t_{mm}$

$$EMAT = t_{TLB} + H * t_{CA} + (1-H) (t_{mm} + t_{CA})$$



Topic : Question

#Q. A processor uses 1-level page table for virtual to physical address translation. Page table is stored in the main memory. Further, the processor has a translation look-aside buffer (TLB), with a hit rate of 96%. The TLB caches recently used virtual page numbers and the corresponding physical page numbers. The processor also has a physically addressed cache with a hit rate of 90%. Main memory access time is 10 ns, cache access time is 2 ns, and TLB access time is also 1 ns.

Assuming that no page faults occur, the average time taken to access a virtual address is approximately?

$$t_{CA} = 2 + 0.1 * 10 = 3 \text{ ns}$$

$$\begin{aligned} E_{MAT} &= 1 + 0.96 * 3 + 0.04 * (10 + 3) \\ &= 4.4 \text{ ns} \end{aligned}$$



2 mins Summary

Topic

Virtual Memory

Topic

Effective Memory Access Time

P₁ PT

0	
1	
2	6

mm

Page

Page 2
of P1

Page 3
of P2



Happy Learning

THANK - YOU

P₂ PT

0	
1	
2	
3	6